CHAPTER 2

Nanoceramics: fabrication, properties, and applications

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2.1 Introduction

Nanoceramics are ceramic materials made up of nanometer-sized grains/crystallites with at least one dimension less than 100 nm. Nanoceramics have excellent mechanical properties, such as greater strength, excellent toughness, and high fatigue resistance [1,2]. The low-stress resistance, low fracture toughness, and poor ductility/machinability of most nanoceramics, however, limit their practical application [3]. The nanoscale creates new mechanical, magnetic, optical, and other properties [4]. The ductility and superplasticity properties of nanoceramics have been discovered and are cost-effective.

The needed characteristics of precursor nanodimensional powders are nanometer size, spherical, uniform, do not form a cluster, and have a high purity level. Due to high packing density, nanopowders are very reactive in solid state during the sintering process, resulting in nanocomposites [5].

Nanosized particles put at grain boundaries of a matrix increase mechanical parameters such as fracture durability and integrity. Nanoceramic composites can have microsized grid grains and nanosized grid grains. Nanoceramic composites have 200% higher strength retention at elevated temperatures and higher creep characteristics than pure ceramics [6].

The excessive use of fossil fuels has led to a deteriorating environment and climate change. The search for low-cost, high-efficiency energy materials is the right solution. Potential uses of nanoceramic materials are in heterogeneous catalysis, solid oxide fuel cells, photoluminescence, dielectrics, and superconductors [7]. Furthermore, nanoceramics are currently widely used in varied fields including electronics, biomedicine, pharmaceuticals, coating materials, nuclear energy, and so on, which are discussed in this chapter [1,8].

2.2 Synthesis of nanoceramics

2.2.1 Sol–gel technique for the synthesis

Particles that are nanosized metal oxides are prepared using this technique. Sol is generated by utilizing inorganic or metal–organic salts. The sol is then hydrolyzed, and

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condensation processes are used to turn it into a gel. The gel is a 3-D structure with many interconnected pores [9]. Finally, any surplus liquid is removed from the gel, which becomes dehydrated, shrinks, and eventually gets converted to the required product. This method allows complicated structures to be formed from a gel phase with a uniform chemical composition without the need for high processing temperatures.

During the synthesis of cationic nanoceramics materials, careful control of hydrolysis and condensation reactions is essential to reduce segregation problems. For synthesis of nanopowders of metal oxide and ceramics, the sol–gel technique is preferred because of its advantages over other techniques [10,11]. The drawbacks of this method are its excessive cost and moisture sensitivity.

2.2.2 Precipitation from solution

A salt precursor is dissolved in water, base solutions are added, and a precipitate of metal hydroxide is obtained in water. After filtering and washing, the hydroxide is calcined to obtain the oxide powder.

Gel precipitation (GPT), precipitation (PPT), and washed precipitation (WPT) can be used [12] for the processing of powder of alumina/ZrO_2 composite; here aqueous ammonia is the precipitating agent. The precipitate is washed with heated water and alcohol before drying in WPT. The precipitate is separated from the liquid before drying in the PPT process. The larger and harder agglomerates are obtained in the GPT approach, whereas smaller and softer agglomerations are obtained in the WPT route. As a result, WPT produces compact agglomerates that sinter well at lower temperatures (Fig. 2.1).

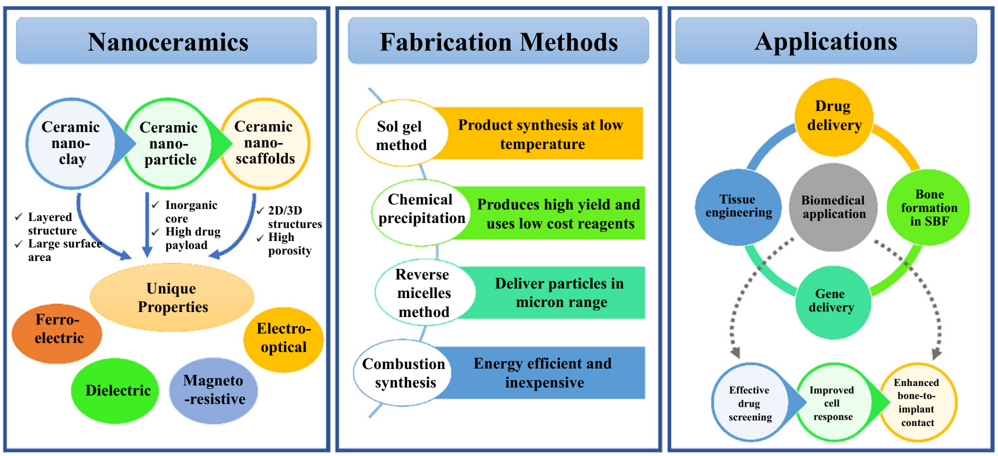

Figure 2.1 An overview of properties, synthesis, and applications of nanoceramics.

2.3 Fabrication techniques

The inclusion of particles of varying sizes provides a higher packaging density. Due to the coarseness of bigger grains, it is difficult to manage microstructural changes during sintering. It also affects densification. Furthermore, when the degree of agglomeration increases, heterogeneous packing occurs, resulting in sintering rate discrepancies and heterogeneous microarrangements. Large defects and lower sintered composite strength can result from a high degree of variability [5]. As a result, fine-sized, freely dispersed powders for both the matrix and reinforcing components are used to enable effective mixing and appropriate particle dispersion in the final powder (Fig. 2.2). The crystalline fraction of the untreated powder can also be used to influence densification and microstructural development. The advantages and disadvantages of various techniques of fabrication can be analyzed.

2.4 Properties of nanoceramics

In contrast to metals, ceramics have their own unique physical, mechanical, and chemical properties. The size of ceramic material has a significant impact on its properties.

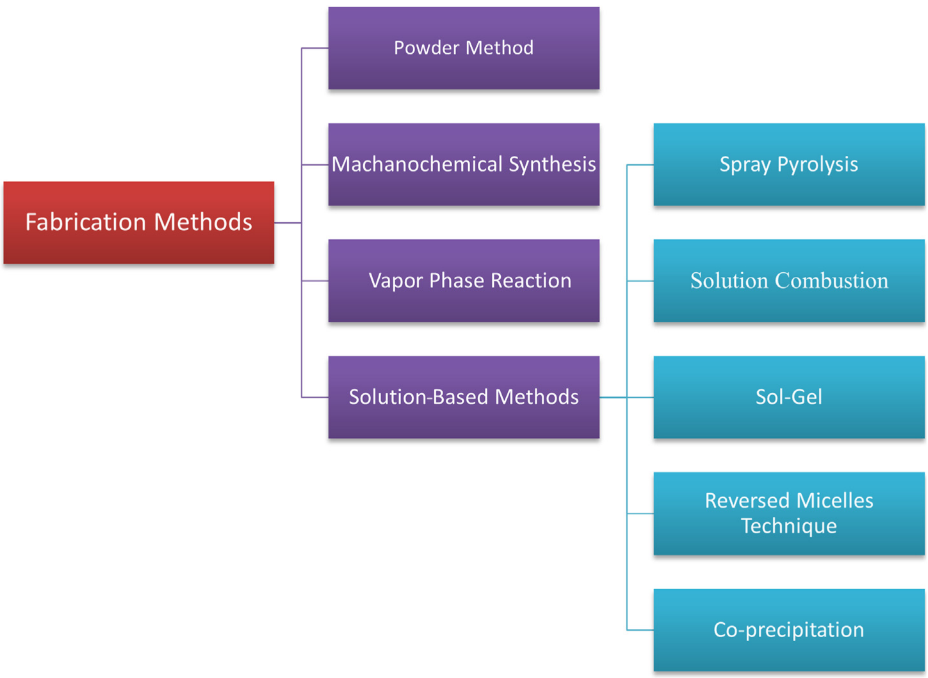

Figure 2.2 Types of fabrication procedures.

2.4.1 Chemical properties

The reactivity [13] of ceramics depends on sites where defects exist. The possibility of getting ceramic powders with excessive porosity [14] makes them suitable for packages. Heterogeneous catalysis and biomedical applications are extremely popular.

2.4.2 Mechanical properties

Ceramics have a high resistance to stress and deformation, which is due to the particle size. Silicon carbides and nitrides, for example, can retain energy at temperatures as high as 1400°C. Some possess metallic strength. Many physical and mechanical properties are affected by particle size [15]. They show useful and scalable mechanical properties like toughness with high Young's modulus.

2.4.3 Sintering properties

Firstly, powders made from individual ceramics are compacted to process nanoceramics from nanophase powders into a crude shape (known as a green body). This compacted powder is then warmed at higher temperatures [16]. Dissemination out of pores (to grain limits) leads to shrinkage of the sample and thus densification occurs. This cycle is known as pressureless sintering. Luckily, nanophase powders were found to be compacted easily. The parameters that need to be maintained during the application of this procedure are shown in Table 2.1.

The sample must be sintered at the lowest temperature feasible for an adequate time to eliminate the porosity and build up stable and well defined grain boundaries. The sintering would finally enhance the effective hardness of the synthesized ceramic.

2.4.4 Ductility and superplasticity

These properties refer to the ability of some polycrystalline materials to go through extensive tensile distortion without necking or breaking. The greatest specialized obstruction in practical applications is ceramic brittleness [17]. Brittle ceramics can be

Table 2.1 The parameters for induction heated sintering process.

Parameter	Applied value
Total power capacity	15 kW
Vacuum level	1×10^{-3} Torr
Resistance heating frequency	50 kHz
Heating rate	100°C–1200°C/min
The output of total power	0%–100%
Applied pressure	10–300 MPa
Cooling rate	500°C/min
Duration time	~10 min

superplastically distorted at modest temperatures. When made from nanoparticles and afterward reheated at higher temperatures high-temperature strengthening is obtained. Brittleness [18] is decreased in ceramics. Ductility [19] and superplasticity tend to be improved.

2.5 Application of nanoceramics

Because of the advantages of the nanoscale, such as the high surface-to-volume ratio, permittivity, porosity, piezo properties, and more, nanoceramics have a diverse and expandable practical scale. Table 2.2 gives a brief overview of the core properties of the nanoceramics for fields of applications and the scalability in certain areas.

2.5.1 Biomedical application

Medical implants made of nanoceramics [53] are gaining popularity because of their excellent properties, like strength, hydrophilicity, and biological compatibility. The present focus is to produce new nanoceramics that are useful for biomedical applications [54]. Cancer treatment, bacterial and viral infection treatment [55], oxygen delivery to damaged tissue, tissue regeneration, drug delivery [56], and gene delivery are some examples.

2.5.2 Simulated body fluid

Surface manipulation and nanoscale coatings are important in the development of substances that are body-interactive and aid in tissue rejuvenation and physiology restoration. Hydroxyapatite layers [56] are produced in simulated body fluid (SBF) on organic or inorganic substrates. Biomaterials show exceptionally fine crystals of carbonate ions that contain apatite after immersion in SBF. Bone cells [57] differentiate and proliferate on this apatite layer, displaying high bioactivity, and the layer facilitates the formation of new bone (as shown in Fig. 2.3).

Under normal conditions, SBF is supersaturated in calcium and phosphate in comparison to apatite. Therefore, if there is an effective functional group in a material for the nucleation of apatite upon its surface, apatite will form spontaneously. The rate of nucleation can be increased if there is an excessive amount of Ca^{2+}, Ti-OH, or the presence of similar groups. This approach may be used to assess an artificial material's bioactivity by looking at its surface's capacity to generate apatite.

2.5.3 Tissue engineering

In tissue engineering, [58] nanoceramics combine with living cells and bioactive scaffolds to deliver essential cells to the sites that are damaged in the body. Stem cells are incorporated into bioceramics. They can reproduce for a longer duration. Adult stem cells can be found in differentiated tissues as pluripotent cells. They regenerate and

Table 2.2 Applications, core qualities that aid them, and the scalable applicative points of nanoceramics.

	Application sector	Types of nanoceramics employed	Unique characteristic	large scale usage	References
Eco-conservation	Pollutant adsorbent for wastewater management	Al_2O_3, SiO_2, ZrO_2, and $Ca_3(PO_4)_2$ γ-Al_2O_3 [79], [80]	Mechanical and chemical resistance, High surface: volume, high liquid permeability with adjustable pore sizes at low pressure.[81]	Removal of bacteria and viruses Removal of dyes Removal of inorganic ions Removal of organic debris Nanofiltration	[20–22] [23,24] [25–27] [28] [2]
Medicine and health	Screening equipment, spectacles, bone tissue repair, guided medication administration, implants, radiation treatment, prosthetics, endoscopy, and orthopedics.	Al_2O_3, SiO_2, ZrO_2, [Al_2O_3 + ZrO_2]-ATZ and ZTA, $Ca_3(PO_4)_2$, TiNbN, Zwitterionic nanoceramics,	Nontoxic, Compatible, elevated bioactivity	Osseous/articulation clinical devices for the hip, shoulder, and more Osteoblast proliferation and Bone construction aids Hip and knee Arthroplasty Knee Prostheses Dentistry	[29–33] [29,34–36] [37,38] [39] [40–43]
Defense and space	Protective ballistics, nuclear fuel pellets, tiles for space shuttles, jet blades, and missile nose cones [79], [108]	Al_2O_3, Si_3N_4, SiC, and AlN, TiO_2	Excellent durability and endurance, exceptional weathering resilience, dimensional consistency, resistance to abrasion, minimal toxic effects, and thermal resilience. [109]	Ballistic helmets Optoelectrical-photoelectrochemical anticorrosion system	[44] [45]
Electronics	Lasers, capacitors, IC, modulators, and microphones.	$BaTiO_3$, lead zirconate titanate films,	High and low (versatile) dielectric coefficient, piezoelectric, permittivity [112], [113]	Capacitors, transformers, and microphones	[7,46–49]
Pharmaceuticals	Drug developmental biochemical research	Carboxymethyl-β-cyclodextrin + Fe_3O_4/SiO_2	Enantioselective environment-stereo selectivity, electric and spin polarization properties physical interactions, and assembly frameworks [118]	Chiral drug separation	[50,51]
Spintronic research	Sensors, detectors, and optoelectronic devices.	[La^{3+} + $BaTiO_3$]	excellent heat stability, strong dielectric constant, and reduced dielectric loss	Pyroelectric, chemical, and biosensors, IR-detectors	[52]

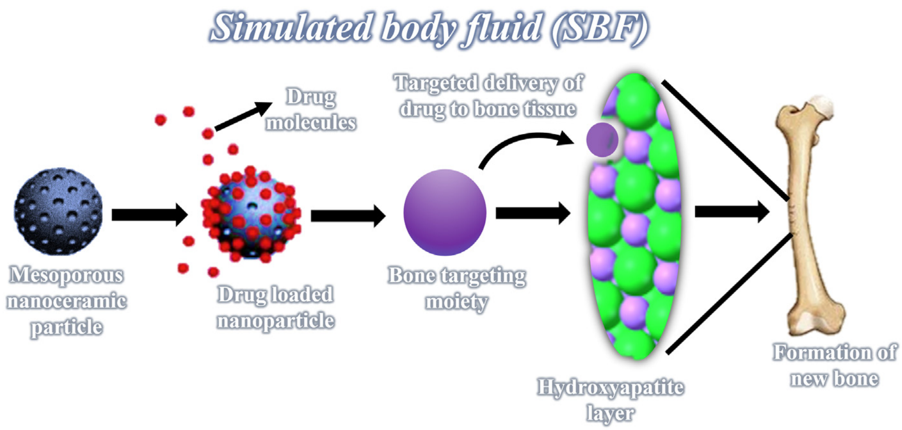

Figure 2.3 Formation of bone on the layer of apatite in SBF. *SBF*, Simulated body fluid.

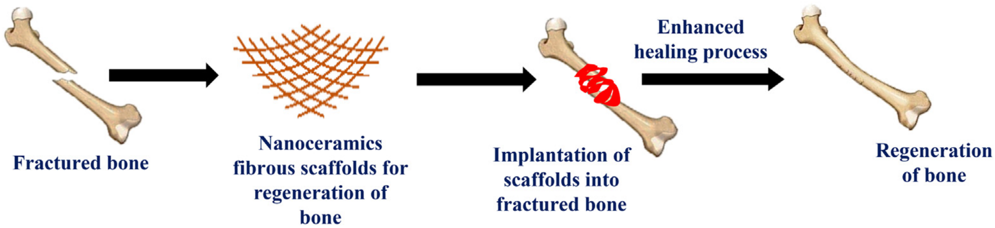

Figure 2.4 Tissue engineering mediated by nanoceramics for bone regeneration.

specialize to produce a certain cell type of tissue from which they emerged [59]. Nanoscale bioactive three-dimensional scaffolds can be useful for the treatment of gaps in long-bone shafts (Fig. 2.4) [60].

Due to their remarkable bioactivity and propensity to attach to hard tissue, Si-based bioactive nanoceramic composites have been developed as appealing materials for bone and tooth regeneration. Different pore diameters are critical for the creation of new tissue in bone tissue regeneration. Micropores operate as mineralization nucleation sites, mesopores as nutrition and developmental component reservoirs, while macrospores cause cell infiltration.

Because of its better potential to induce bone tissue regeneration while also acting as a drug delivery carrier, biologically active glass is rapidly being studied for use in the manufacture of bone-repair scaffolds. By altering the surface topography or the morphology of scaffolds and transplants, clay-based nanoceramics are being employed as coatings to increase the biological effect and incorporation (Fig. 2.5) [61].

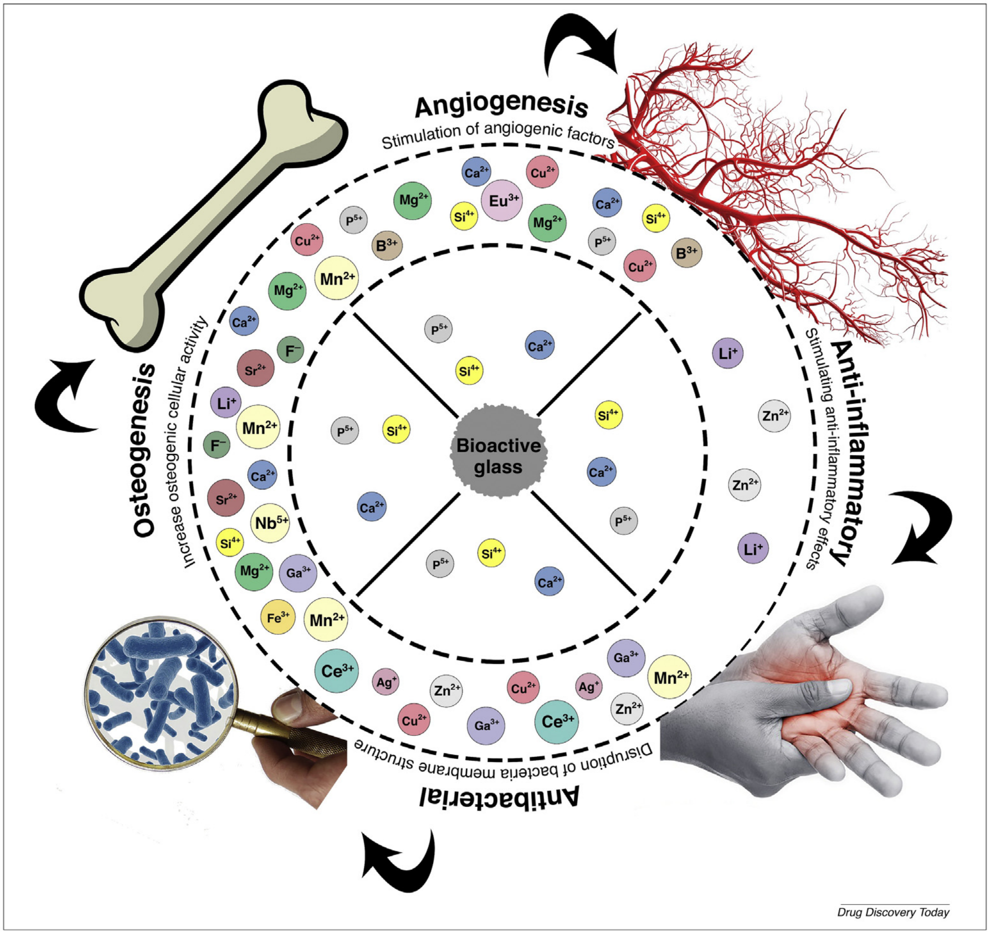

Figure 2.5 Representation of applications of nanoceramic (bioactive glass) in the therapeutic and biomedical field. Source*: From S. Kargozar, F. Baino, S. Hamzehlou, R. G. Hill, M. Mozafari, Bioactive glasses entering the mainstream, Drug Discovery Today, 23 (10) (2018) 1700–1704. https://doi.org/10.1016/j.drudis.2018.05.027*.

2.5.4 Drug delivery

Ceramic nanoparticles are formed of substances that lay between metals and nonmetals in terms of properties. They have poor electrical and thermal conductance, as well as a strong elastic modulus, stiffness, and resistance to corrosion [62]. Nanoceramics have also been made using a combination of these materials to improve biocompatibility, drug release mechanisms, therapeutic impact, and other features. Mesoporous silica nanoceramics are one of the most potent pharmaceutical administration methods in anticancer medication. The large surface area, configurable pore diameter, customizable particle size and shape, and exceptional biocompatibility are just a few of the advantages of this material over others. Mesoporous silica nanoparticles can deliver both hydrophobic and hydrophilic medicinal molecules with precision. They have also been used for therapeutic and diagnostic applications, such as visualizing the delivery location. Functionalizing these particles with chemicals like folic acid and VEGF-121, or lactose amination of the surface provides an effective targeting capability. Combining silica with thermally responsive polymers has also resulted in pH-sensitive, temperature, and magnetic field-sensitive drug delivery systems [62,63]. Table 2.3 shows the application of silica nanoceramics in anticancer drug delivery.

Table 2.3 Silica nanoceramics used in anticancer drug delivery.

Nanoceramic product	Challenges	Inference	References
Doxorubicin, a mesoporous silica-coated layer by layer	DOX has harmful effects due to uncontrolled and untargeted delivery	This nanosystem was designed to respond to acidic environments in a certain way (endosomal compartments of cancer cells). It has great stimuli-responsive drug release capability, excellent biostability, and minimal premature drug leakage at pH 7.4. Experiments in vivo revealed an extremely efficient tumor growth inhibition rate.	[64]
Charge reversal, pH-responsive polymer-coated doxorubicin-loaded with mesoporous silica nanoceramic	It has toxicity due to nontargeted delivery	Nanoceramics were made with 3-aminopropyl-triethoxysilane. DOX was effectively transported and released to the nucleus of HeLa cells using nanoceramics.	[57]
Lactosaminated mesoporous SiO_2 nanoceramics filled with docetaxel for asialoglycoprotein receptor targeting.	Systemic adverse effects of drug	HepG2 and SMMC7721 cells were effectively inhibited by Lactosaminated mesoporous silica nanoceramics loaded with docetaxel.	[65]
Cisplatin-loaded mesoporous silica nanoparticles with an avidin cap	Systemic harmful effects of cisplatin due to nontargeted delivery	Release of cisplatin via MMP9, which resulted in apoptotic cell death restricted to the lung tumor locations. MMP9 responsive nanoceramics also enabled excellent cisplatin and proteasome inhibitor combination drug delivery.	[66]

(*Continued*)

Table 2.3 (Continued)

Nanoceramic product	Challenges	Inference	References
Mesoporous silica nanoceramics loaded with conjugated-sunitinib	Insufficient medication administration limits therapeutic efficacy.	Due to effective targeting, Sunitinib was supplied to U87MG tumor cells in larger amounts.	[67]
Cisplatin, mesoporous hollow silica nanospheres	Application is limited to traditional dosage form	Cisplatin, folic acid, and rhodamine isothiocyanate were coupled to the external interface, which was acid-functionalized. Encapsulated superparamagnetic $CoFe_2O_4$ nanoparticles and a hydrophobic anticancer agent Nanospheres showed improved cytotoxicity, as well as MRI contrast characteristics, as compared to individual drugs.	[68]

2.5.5 Nanoceramics as disinfectant agents for water and wastewater purification

Waterborne bacterial infections are linked to various illness outbreaks recorded around the world because of contaminated water, creating a substantial hazard to the health-care system. Solar treatment, UV treatment, catalytic oxidation, heat disinfection, and advanced oxidation processes have all been investigated as alternatives to chemical disinfection of wastewater. Despite their apparent effectiveness, each of these drinking water treatment technologies has drawbacks that limit or prevent their usage in specific situations. Certain negatively strained bacteria, for example, may need longer treatment for complete decontamination by solar water disinfection method; due to bacterial regeneration documented after a day of the treatment because of initiation of the DNA repair process [69,70].

Magnetic nanostructures with well-controlled dimensions and components have received interest from researchers due to their successful implementation in versatile procedures as a secure alternative for the potential threat of bacterial infections. Due to their structural features, quasicrystal and ceramic nanoparticles have shown promising properties among the produced nanoparticles. Nano-porous Co_2O_3/Cu_2O_3 with alumina and silica was found to be the most efficient composite in removing Gram-positive and Gram-negative bacteria from water in recent research, showing that it is a harmless, environment-friendly, and cheap alternate disinfectant for water supply and

Table 2.4 The nanomaterials and their corrosion resistance capabilities.

Material	Nanomaterial coating	Coating thickness	Substrate	Corrosion resistance	References
Zirconia oxides	Thin films of ZrO_2	50–350 nm	Preoxidized 304L stainless steel	Best when the surface is oxidized before coating	[78]
		500 nm	Alumina-silica refractory material	Corrosion loss decreases by 18%	[79]
		0.5–0.8 μm	316L stainless steel	Better with temperature aid 400°C–650°C	[80]
Titanium oxides		TiO_2 NP	550 nm	Greatest for coating thickness 464 nm	[81]
		TiO_2 thin films	370 nm	Greatest with The n-modified surface of tio_2	[82]
Aluminum Oxides	Al_2O_3 thin film	200 nm	X40crmov5–1 steel	Greatest with deposition temperature 300°C	[83]
		50 nm	Carbon steel	Unable to protect the steel	[84]
		10, 20, 50 nm	Copper	10 nm is superior to 50 nm	[85]
Nickel alloys	SiO_2–Ni–P	29 μm	API-5L X65 steel substrates	Optimal conc. At 2 wt.% SiC	[86]
	SiC–Ni–P	19–21 μm	St37 tool steel substrate	4 wt.%	[87]

wastewater treatment. As a result, highly porous magnetic nanomaterials could be regarded as a new nanotechnique for disinfecting wastewaters. The doped Co_3O_4 nanoceramics materials have the potential to be utilized in sensing devices as well [20].

2.5.6 Corrosion-resistant coatings

Nanoceramic coatings are beneficial for solving corrosion problems in every electrode and material body [71,72]. Types of nanoceramic [73] coating include thermal spraying, plasma spraying [74], sputter coating, and electrochemical coating [24]. The ceramic films range from 50 nm to a few μm in thickness. Elevated-temperature resistible structural materials are Si_3N_4, diamond-like coatings, silicon carbide [75], and boron nitride [76]. They have magnificent thermomechanical properties [72]. The weathering potential of coated substrates is positive, but the corrosion current density decreases. The rate of corrosion slows down. Some materials and their abilities in corrosion resistance are shown in Table 2.4.

2.6 Conclusion and prospects

The development and vivid research of nanoceramics have increased ceramic material performance in various applications utilizing their chemical, mechanical, electrical, and other characteristics. These nanoceramics synthesized and fabricated in film-like forms have allowed tremendous progress in simplifying the daily life of producers and customers of various products for different sectors.

Ceramic processing has expanded on an industrial scale in fields such as medicine, pollution control, automotive durability and shielding, the aircraft industry, or improving therapeutic practices not only as an alternative commodity but as a superior need. Nanoceramic coatings have critical features that could lead to a revolution in the upcoming implantable devices for bone regeneration. It has been demonstrated that the implantation need not have to be balanced to the internal milieu of the patient's body where bioactivity is sought for bone implants without the use of an expensive graft and thus revolutionizing the medical field. Due to the widespread adaptability, this field has caught the attention of the whole science establishment and guarantees a lot of technology in the coming years. As a result, scaling for these nanoceramics might be a challenging task.

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